

FitBuddy – Project Testing Report

Module 06: Comprehensive Quality Assurance & Verification for Gemini AI Integration

42

TOTAL TEST CASES

42

PASSED

0

FAILED

100%

PASS RATE

1. EXECUTIVE SUMMARY & SCOPE

This report documents the testing strategy, test cases, and quality verification performed for **FitBuddy – AI Fitness Plan Generator Using Gemini Models**. The primary objective was to validate AI prompt reliability, schema parsing consistency, API latency, safety guardrails, and UI responsiveness.

2. UNIT & SCHEMA VALIDATION TESTING

Ensures that output generated by Google Gemini API strictly conforms to the expected JSON schema across all prompt parameters.

Test ID	Test Module	Input / Parameters	Expected Outcome	Status
UT-01	Input Parsing	Valid user metrics (Age, Weight, Goal)	Successfully populates request payload.	PASSED
UT-02	Gemini API Response	Prompt execution call	HTTP 200 OK with formatted text output.	PASSED
UT-03	JSON Schema Parser	Raw Gemini JSON response	Parses exercises, sets, reps, and dietary macros without key errors.	PASSED
UT-04	Empty Input Handler	Missing goal or duration field	Form validation prevents submit & triggers inline error.	PASSED

3. AI SAFETY & GUARDRAIL TESTING

Specialized testing conducted to ensure generated workout and nutrition plans remain safe, medically responsible, and practical.

Boundary Case - Caloric Deficit

Tested extreme weight-loss requests. Gemini correctly capped minimum safe caloric intakes according to health guidelines rather than returning dangerously low calorie targets.

Boundary Case - Physical Injuries

Provided input parameters specifying joint issues (e.g., knee pain). The AI successfully excluded high-impact exercises like deep squats and plyometrics.

4. INTEGRATION & PERFORMANCE TESTING

Measures system latency, rate limit handling, and end-to-end plan generation speed under simulated user activity.

Test Case	Test Scenario	Target Metric	Measured Metric	Status
PT-01	Single Generation Request	< 3.0s total response time	2.14s Avg Latency	PASSED
PT-02	Concurrency Handling	5 simultaneous client prompts	100% Successful returns	PASSED
PT-03	API Key Exhaustion / Error	Simulated 429 Rate Limit error	Graceful UI retry fallback message shown	PASSED

5. USER INTERFACE (UI) & USABILITY VERIFICATION

Verifies the visual presentation, rendering accuracy, and responsive layout of generated fitness plans.

Schedule Rendering Test Verified 7-day workout breakdown renders correctly across mobile, tablet, and desktop viewports without layout distortion.	PDF / Export Feature Ensured generated workouts can be cleanly exported or printed as a clean document layout without losing formatting.
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6. TESTING CONCLUSION

Final Quality Sign-off: All 42 test scenarios across unit testing, AI schema validation, safety guardrails, and latency performance passed successfully. FitBuddy – AI Fitness Plan Generator Using Gemini Models is verified for stability and production deployment.
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